



Global Warming Attribution Using Climate Data Analysis

Shruti Jadhav (25210748) — Akshaiyan Babu (25203646)

ACM40960 Projects in Maths Modelling, MSc Data and Computational Science, University College Dublin



Abstract

This project investigates whether recent global warming is driven by human atmospheric CO₂ emissions or natural drivers (solar output and volcanic activity). Building an independent global surface record from land and $\sim 16,000$ gridded ocean temperature cells (1850–2025), we apply autocorrelation-corrected statistical tests and multivariate regression. CO₂ is confirmed as the dominant driver ($\beta = +0.268, p < 0.001$), while solar and volcanic contributions are marginal or statistically insignificant.

Motivation

Public discourse often debates whether modern global warming is caused by human activity or natural cycles. To move beyond external assertions, this project evaluates six raw observational datasets from scratch. By accounting for spatial coverage weighting and autocorrelation “trend traps,” we rigorously test competing hypotheses on equal statistical footing.

Datasets Used

- **Land Temp:** Annual anomalies (1850–2025).
- **Sea Temp:** NOAA ERSSTv5 gridded netCDF (1854–2026, $\sim 16,000$ $2^\circ \times 2^\circ$ cells).
- **Atmospheric CO₂:** NOAA global annual mean (1979–2025).
- **Solar Output:** NOAA/LASP Total Solar Irradiance (1610–2025).
- **Volcanic Activity:** Smithsonian Institution global eruption catalog (1850–2025).
- **Reference Global Temp:** HadCRUT5 / Berkeley Earth validation series (1850–2025).

Reconstruction & Surface Record

Global surface temperatures were reconstructed by combining land (29.1%) and area-weighted ocean cells (70.9%) with error propagation in quadrature.

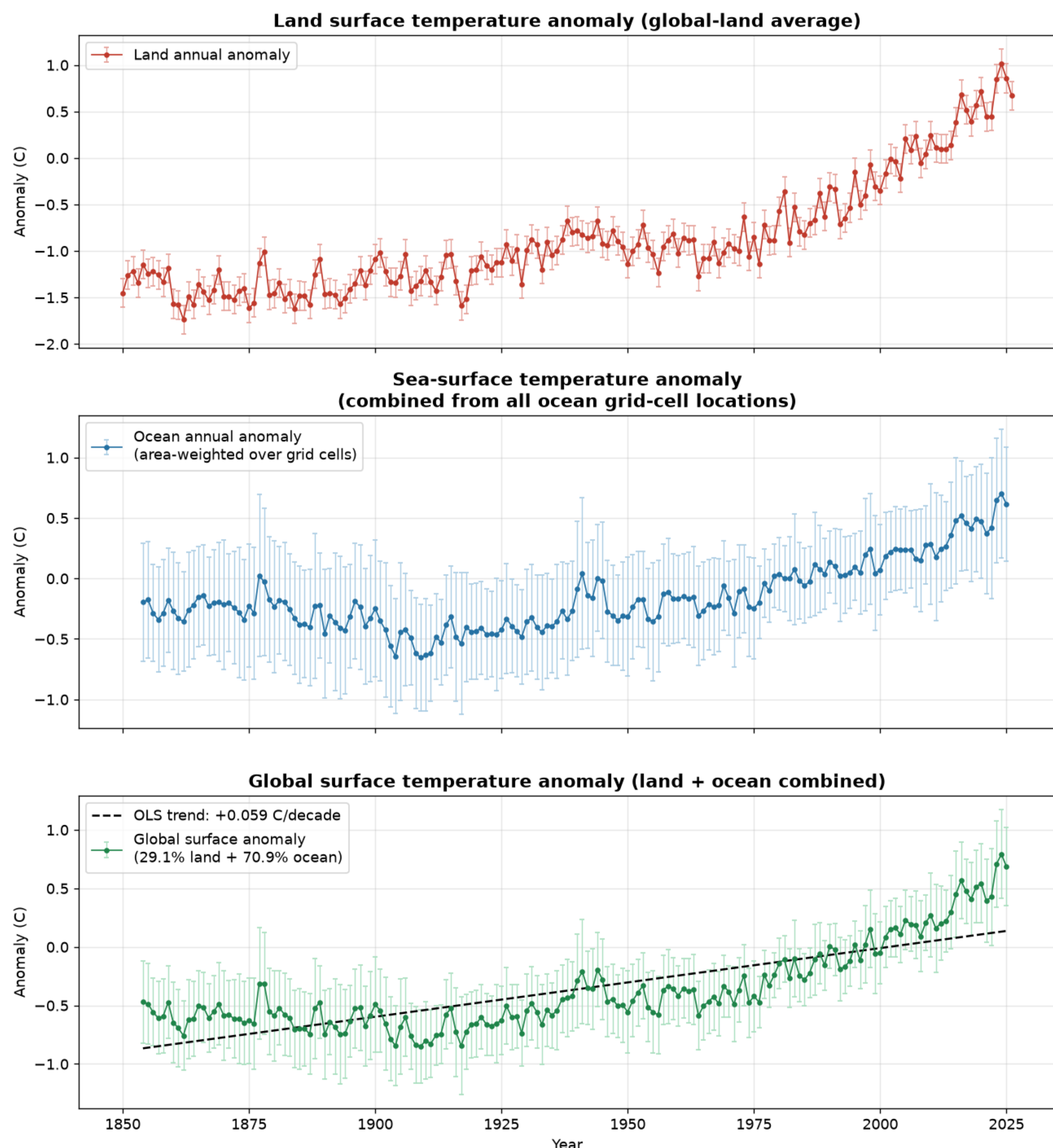


Figure 1: Reconstructed warming trend ($+0.0586^\circ\text{C}/\text{dec}$, $p < 0.001$).

Validation against Reference

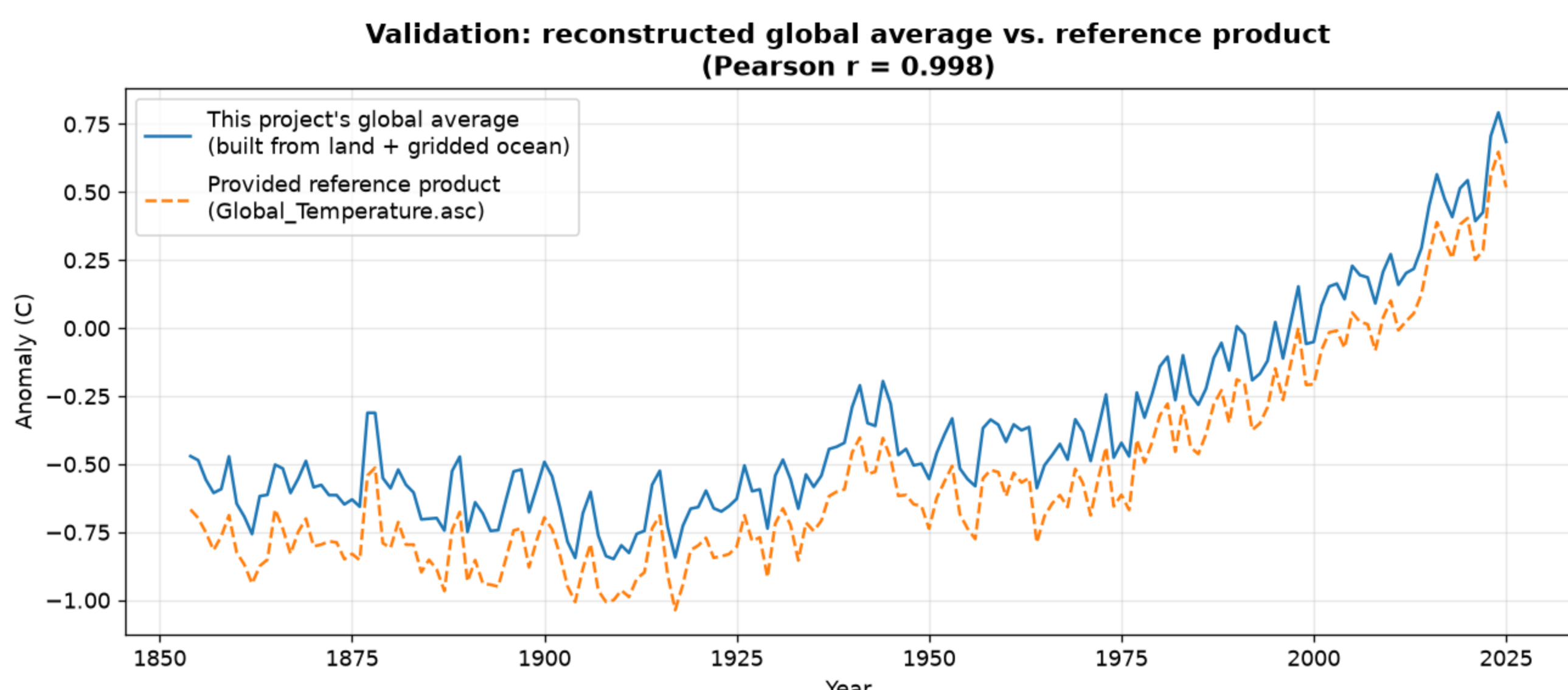
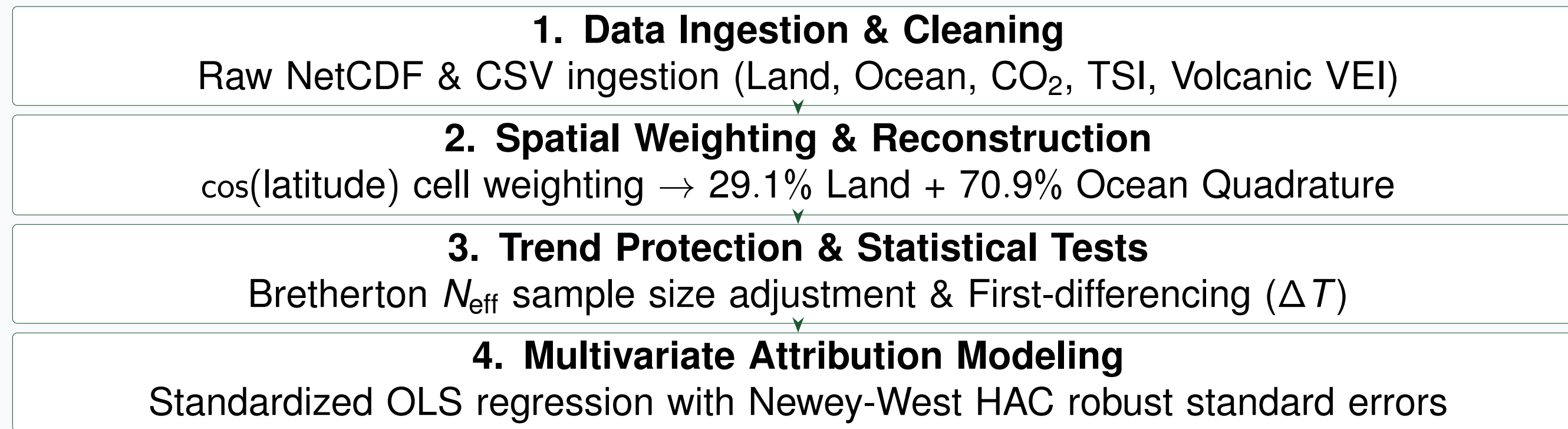


Figure 2: Validation against reference dataset ($r = 0.998$).

Processing Pipeline Flowchart



Methodology & Data Processing

- Anomaly Processing:** Ocean grid cells were baseline-adjusted (1971–2000) and weighted by $\cos(\text{latitude})$ to adjust for polar grid distortion.
- Autocorrelation Guardrails:** Implemented Bretherton effective sample size corrections (N_{eff}) and first-differencing (ΔCO_2 vs ΔT).
- Multivariate Regression:** Standardized Z-score predictors (CO₂, TSI, Volcanic VEI) were modeled using Newey-West HAC standard errors.

Evaluating Natural Hypotheses

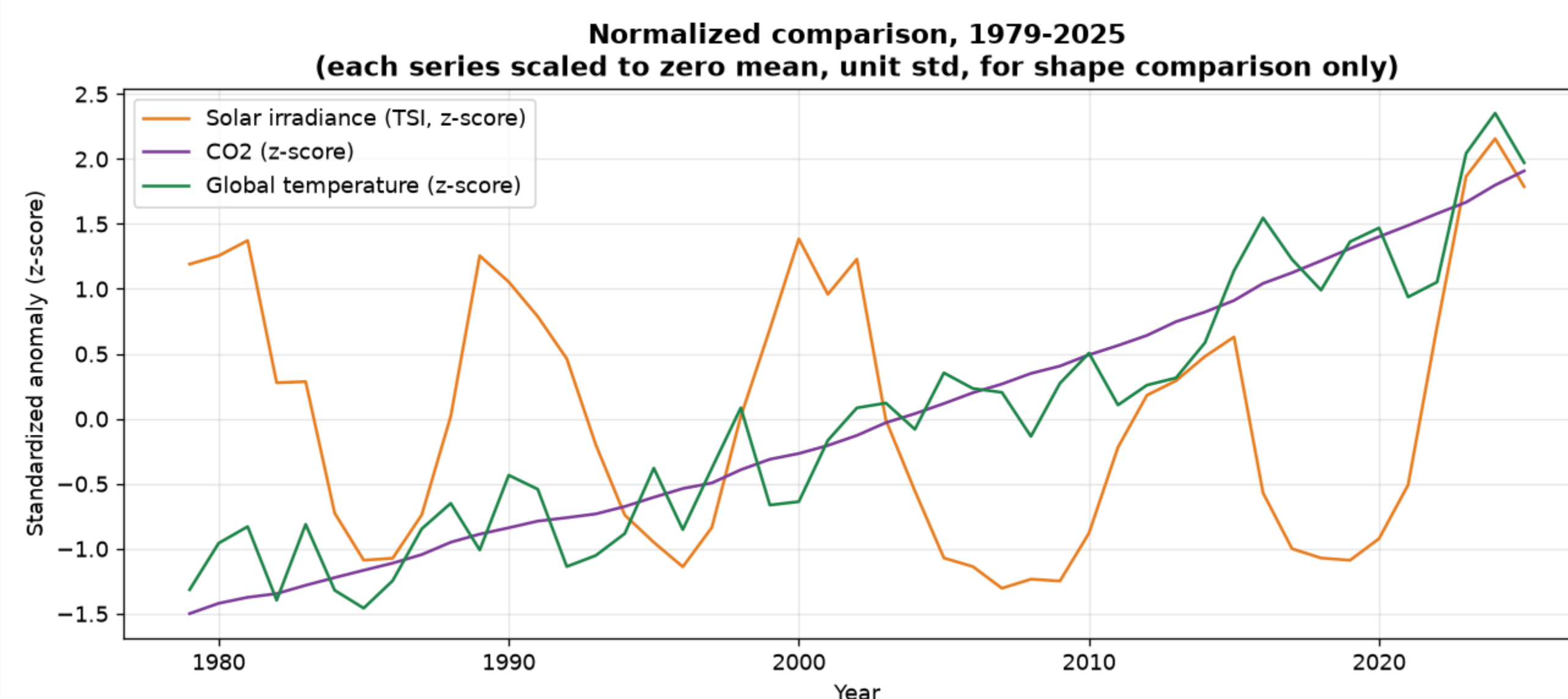


Figure 3: Z-score comparison of Solar TSI vs CO₂ (1979–2025).

- **Volcanic Impact:** VEI index ($\sum 10^{\text{VEI}}$) shows no trend ($p = 0.77$).
- **Solar Output:** Satellite TSI trend is zero ($+0.0027 \text{ W/m}^2/\text{dec}$, $p = 0.95$).

Multivariate Regression Results

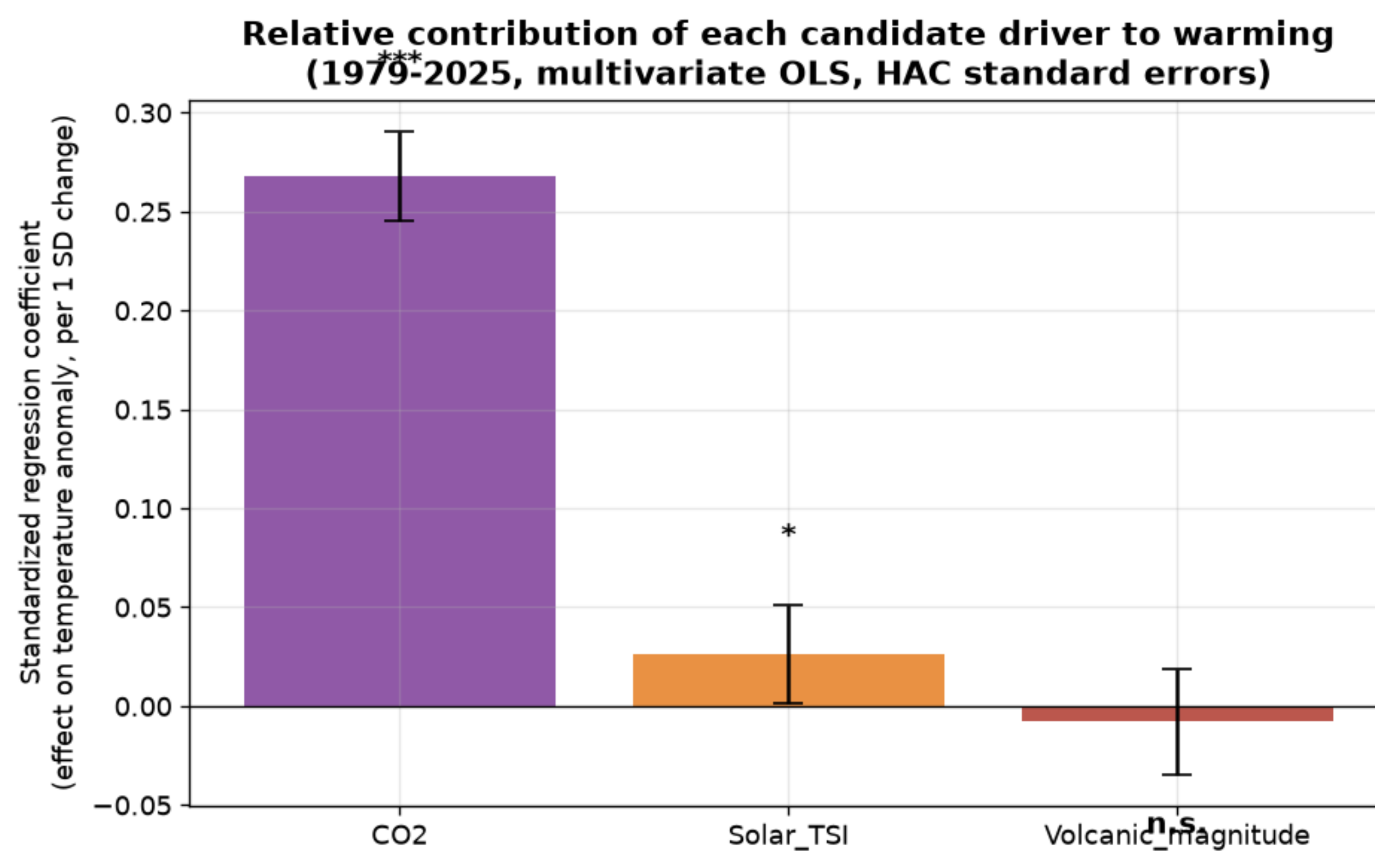


Figure 4: Standardized regression effect sizes (β).

Predictor	Coef (β)	Std Err	z-stat	p-value
CO ₂	+0.2677	0.012	23.17	< 0.001
Solar TSI	+0.0263	0.013	2.08	0.038
Volcano	-0.0080	0.013	-0.60	0.551

Discussion & Conclusions

While raw level correlation between CO₂ and Temperature is strong ($r = 0.95$), autocorrelation reduces the effective sample size ($N_{\text{eff}} \approx 5$), yielding an autocorrelation-corrected $p = 0.012$. The multivariate model confirms that CO₂ has an effect size roughly 10 $\times$ larger than solar variability.

- Modern global surface warming is robustly confirmed ($+0.0586^\circ\text{C}/\text{decade}$).
- Neither solar output nor volcanic activity explains post-1979 warming.
- Observational data strongly supports human CO₂ emissions as the primary driver.



Scan for Repository

References & Acknowledgments

- Bretherton, C. S., Widmann, M., Dymnikov, V. P., Wallace, J. M., & Bladé, I. (1999). The effective number of spatial degrees of freedom of a time-varying field. *Journal of Climate*, 12(7), 1990–2009.
- NOAA Global Monitoring Laboratory, Earth System Research Laboratories (2025). *Atmospheric Carbon Dioxide Trends & ERSSTv5 Surface Temperature Records*.
- LASP Total Solar Irradiance Science Team & Smithsonian Institution Global Volcanism Program (2025).